CHAPTER 4 - EXTENDING AND REFINING LM / GLM

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In this chapter, we study how to incorporate structured information in a GLM specification, as well as procedures to refine the model, performing model and variable selection.

1 MIXED MODELS

A first natural extension of linear models and GLMs can be obtained considering mixed models. In a mixed environment, we observe covariates but we also have access to a grouping information of our observations. Those (known) groups are already providing a fixed partition of observations, whereas within each group of data we can expect the quantity we observe to be more homogeneous. Even if each observation group assignment can be viewed as a covariate as well, we can accommodate for some covariate effects that depends on the specific group.

Example 1.1. Suppose, for example, we observe the gestational time for different woman, sampled from several hospitals. We want to model the gestational time as function of some covariates we measure, such as life habits (e.g. smoker or not, drinks alcohol or not, etc) and biometric measures. The hospitals are spread across the country, with different climates and habits, and it is reasonable to assume that they can eventually differ in the respective covariate effects.

The idea of mixed model is to consider some of the regression coefficients to be group-specific, that can vary group by group. Luckily, we can easily deal with this family of model in a Bayesian setting, by suitably choosing an appropriate prior distribution. Hence, we extend linear models considering covariate effects that can possibly be group-specific.

We consider a set of $\mathbb{Y}$-valued response variables $y_1, \dots, y_n$, with $\mathbb{Y} \subseteq \mathbb{R}$, and $\mathbb{X}$-valued covariates $\{z_{i,1}, \dots, z_{i,p}\}$, with $\mathbb{X} \subseteq \mathbb{R}^p$, for $i = 1, \dots, n$. Here, data are divided into $k \geq 1$ groups. Hence, we also observe a sequence of allocation variables $\{c_1, \dots, c_n\}$, where for the generic c_i we have that $c_i = j$ if the ith observation belongs to group j. We denote by

- $\boldsymbol{x}_i = (x_{i,1}, \dots, x_{i,p})^\intercal$ the covariates associated to fixed effects, whose regression coefficients are given by the vector $\boldsymbol{\beta}$, out of $\boldsymbol{z}_i$ and after possible transformation;
- $\boldsymbol{u}_i = (u_{i,1}, \dots, u_{i,q})$ denote the covariates with random effects, i.e. group-specific effects, $\boldsymbol{\gamma}$, out of $\boldsymbol{z}_i$ and after possible transformation.

Note that both $\boldsymbol{x}_i$ and $\boldsymbol{u}_i$ are subsets of $\{1, \boldsymbol{z}_i\}$, possibly with overlaps, i.e. a covariate can have a fixed and a random effect at the same time, where the fixed effect describe a common behavior shared across groups and the random effect a group-specific impact of the covariate on the response variable.

A first model specification, for linear regression mixed models, is given by

$$y_i = \boldsymbol{x}_i^\intercal \boldsymbol{\beta} + \boldsymbol{u}_i^\intercal \boldsymbol{\gamma}_{c_i} + \epsilon_i,$$

where $\epsilon_i \sim N(0,1)$ are independent and identically distributed error terms, as usual, for $i = 1, \dots, n$. The same model can be expressed in matrix notation,

$$\boldsymbol{y} = \mathrm{X}\boldsymbol{\beta} + \mathrm{U}\boldsymbol{\gamma}_{c_i} + \boldsymbol{\epsilon},$$

with X being a $n \times p$ matrix whose ith row is given by $\boldsymbol{x}_i^\intercal$, U a $n \times q$ matrix with ith row equal to $\boldsymbol{u}_i^\intercal$, and $\boldsymbol{\epsilon} \sim N(\mathbf{0}, \sigma^2 \mathrm{I}_n)$. It clearly reminds the structure of a linear model, but now with two additive terms modelling the common effects and the group specific effects.

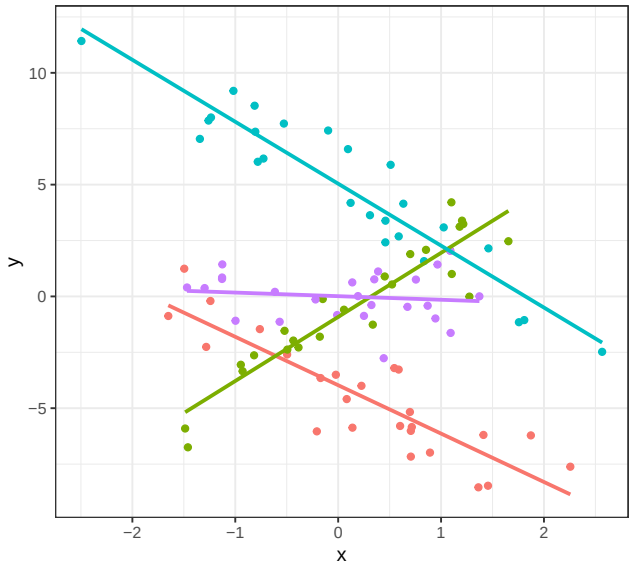

Figure 1: Graphical representation of a simple linear regression model, with random slope and intercept.

We can appreciate how grouping the data helps to understand and explain relationships between the response variable and the covariates.Think about modelling the whole data without knowing the groups, with a set of data as in Figure 1 would definitely lead to miss-interpreting the covariate effects on the response variable.
With mixed model, we can cover different specifications depending on which quantity we consider to be group-dependent, which usually depends on the inferential question we want to investigate. Some remarkable cases are the followings.

- If there are no covariates in $\boldsymbol{u}_i$, i.e. is an empty set, or $\boldsymbol{\gamma}_j = \mathbf{0}$, for any $j = 1, \dots, k$, the model collapse on a linear model with covariates X and regression coefficients $\boldsymbol{\beta}$.

- If $\boldsymbol{u}$ contains only the intercept term, e.g.

  $$\boldsymbol{u}_i = 1, \qquad \boldsymbol{\gamma}_j = \gamma_j,$$

  the model is a linear model with random intercept. We assume there is a systematic group effect on the response variable, but no differences in other regression coefficients.

Note that, as for linear models, we require X and U to be both full rank matrices. Furthermore, the prior distribution of the random effect should be centered in $\mathbf{0}$, otherwise the model is not identifiable.

Definition 1.2 (Identifiability)**.** A generic model $f(\cdot \mid \theta)$, $\theta \in \Theta$ is said to be identifiable if

$$f(\cdot \mid \theta_1) = f(\cdot \mid \theta_2) \qquad \Rightarrow \qquad \theta_1 = \theta_2, \qquad \text{for any } \theta_1, \theta_2 \in \Theta.$$

For the linear regression mixed model case, the likelihood equals

$$\mathrm{L}(y_{1:n} \mid \boldsymbol{x}_{1:n}, \boldsymbol{u}_{1:n}, c_{1:n}, \boldsymbol{\beta}, \boldsymbol{\gamma}) = (2\pi)^{-\frac{n}{2}}(\sigma^2)^{-\frac{n}{2}} \exp\left\{-\sum_{i=1}^{n} \frac{1}{2\sigma^2}(y_i - \boldsymbol{x}_i^\intercal\boldsymbol{\beta} - \boldsymbol{u}_i^\intercal\boldsymbol{\gamma}_{c_i})^2\right\}.$$

We recognise an expression similar to the linear regression model one, with an extra term. Hence, we consider a similar prior distribution. The model specification is completed by setting the following distributional assumptions.

$$\begin{aligned} Y_i \mid \boldsymbol{x}_i, \boldsymbol{u}_i, c_i, \boldsymbol{\beta}, \boldsymbol{\gamma}_{c_i} &= N(\boldsymbol{x}_i^\intercal\boldsymbol{\beta} + \boldsymbol{u}_i^\intercal\boldsymbol{\gamma}_{c_i}, \sigma^2), \qquad i = 1, \ldots, n, \\ \boldsymbol{\beta} &\sim N(\boldsymbol{b}_0, \Sigma_0), \\ \boldsymbol{\gamma}_j &\sim N(\boldsymbol{\tau}_0, \Phi_0), \qquad j = 1, \ldots, k, \\ \sigma^2 &\sim IG(a_0, b_0). \end{aligned}$$

We first note that the prior specification is not joint anymore, like in the linear regression model of the previous chapter. Here, we assume an independent prior specification, which hopefully leads to marginal tractability. As usual, the prior distributions for fixed and random effect should be dispersed enough over their support, i.e. Σ_0 and Φ diagonals should be large enough. Further, the previous specification assumes independent priors on the parameters of interest. We can write explicitly the full conditional distributions for this specification.

Proposition 1.3. *The independent prior distribution specification leads to marginal conjugacy. Specifically, a posteriori we have*

$$\begin{aligned} \boldsymbol{\beta} \mid y_{1:n}, \boldsymbol{x}_{1:n}, \boldsymbol{u}_{1:n}, c_{1:n}, \boldsymbol{\gamma}_{1:k}, \sigma^2 &\sim N(\boldsymbol{b}_n, \Sigma_n), \\ \boldsymbol{\gamma}_j \mid y_{1:n}, \boldsymbol{x}_{1:n}, \boldsymbol{u}_{1:n}, c_{1:n}, \boldsymbol{\beta}, \sigma^2 &\sim N(\boldsymbol{\tau}_n, \Psi_n), \qquad j = 1, \ldots, k, \\ \sigma^2 \mid y_{1:n}, \boldsymbol{x}_{1:n}, \boldsymbol{u}_{1:n}, c_{1:n}, \boldsymbol{\beta}, \boldsymbol{\gamma}_{1:k} &\sim IG(a_n, b_n), \end{aligned}$$

where

$$\begin{aligned} \boldsymbol{b}_n &= \Sigma_n\left(\Sigma_0^{-1}\boldsymbol{b}_0 + \frac{\mathrm{X}^\intercal\boldsymbol{y}^\gamma}{\sigma^2}\right), \qquad \Sigma_n = \left(\Sigma_0^{-1} + \frac{\mathrm{X}^\intercal\mathrm{X}}{\sigma^2}\right)^\intercal, \\ \boldsymbol{\tau}_n &= \Psi_n\left(\Psi_0^{-1}\boldsymbol{\tau}_0 + \frac{\mathrm{U}_j^\beta\boldsymbol{y}_j^\beta}{\sigma^2}\right)^{-1}, \qquad \Psi_n = \left(\Psi_0^{-1} + \frac{\mathrm{U}_j^\intercal\mathrm{U}_j}{\sigma^2}\right)^{-1}, \\ a_n &= a_0 + \frac{n}{2}, \qquad b_n = b_0 + \frac{1}{2}\sum_{i=1}^{n}(y_i - \boldsymbol{x}_i^\intercal\boldsymbol{\beta} - \boldsymbol{u}_i^\intercal\boldsymbol{\gamma}_{c_i})^2, \end{aligned}$$

with $y_i^\gamma = y_i - \boldsymbol{u}_i^\intercal\boldsymbol{\gamma}$, $\boldsymbol{y}_j^\beta = \{y_i - \boldsymbol{x}_i^\intercal\boldsymbol{\beta} : c_i = j\}$, *and*

$$\mathrm{U}_j^\beta = \left\{\begin{bmatrix} \boldsymbol{u}_1^\intercal \\ \boldsymbol{u}_2^\intercal \\ \vdots \\ \boldsymbol{u}_{n_j}^\intercal \end{bmatrix} : c_i = j\right\}.$$

Proof. First, note that the likelihood function can be factorize as

$$\mathrm{L}(y_{1:n} \mid \boldsymbol{x}_{1:n}, \boldsymbol{u}_{1:n}, \boldsymbol{\beta}, \boldsymbol{\gamma}) = \prod_{j=1}^{k}(2\pi)^{-\frac{n_j}{2}}(\sigma^2)^{-\frac{n_j}{2}} \exp\left\{-\sum_{i:c_i=j}\frac{1}{2\sigma^2}(y_i - \boldsymbol{x}_i^\intercal\boldsymbol{\beta} - \boldsymbol{u}_i^\intercal\boldsymbol{\gamma}_j)^2\right\}.$$

Then, we have for the dispersion term

$$\begin{aligned}
&\pi(\sigma^2 \mid y_{1:n}, \boldsymbol{x}_{1:n}, \boldsymbol{u}_{1:n}, c_{1:n}, \boldsymbol{\beta}, \boldsymbol{\gamma}_{1:k}) \\
&\quad \propto \mathrm{L}(y_{1:n} \mid \boldsymbol{x}_{1:n}, \boldsymbol{u}_{1:n}, c_{1:n}, \boldsymbol{\beta}, \boldsymbol{\gamma})\pi(\sigma^2) \\
&\quad \propto (\sigma^2)^{-n/2}(\sigma^2)^{-a_0-1} \exp\left\{-\frac{1}{\sigma^2}\left[b_0 + \frac{1}{2}\sum_{j=1}^{k}\sum_{i:c_i=j}(y_i - \boldsymbol{x}_i^\intercal\boldsymbol{\beta} - \boldsymbol{u}_i^\intercal\boldsymbol{\gamma})^2\right]\right\}
\end{aligned}$$

which identifies an inverse-gamma distribution, $\sigma^2 \mid y_{1:n}, \boldsymbol{x}_{1:n}, \boldsymbol{u}_{1:n}, c_{1:n}, \boldsymbol{\beta}, \boldsymbol{\gamma}_{1:k} \sim IG(a_n, b_n)$, with

$$a_n = a_0 + \frac{n}{2}, \qquad b_n = b_0 + \frac{1}{2}\sum_{j=1}^{k}\sum_{i:c_i=j}(y_i - \boldsymbol{x}_i^\intercal\boldsymbol{\beta} - \boldsymbol{u}_i^\intercal\boldsymbol{\gamma})^2.$$

Regarding the regression coefficients, we have

$$\begin{aligned}
&\pi(\boldsymbol{\beta} \mid y_{1:n}, \boldsymbol{x}_{1:n}, \boldsymbol{u}_{1:n}, c_{1:n}, \boldsymbol{\gamma}_{1:k}, \sigma^2) \\
&\quad \propto \mathrm{L}(y_{1:n} \mid \boldsymbol{x}_{1:n}, \boldsymbol{u}_{1:n}, c_{1:n}, \boldsymbol{\beta}, \boldsymbol{\gamma})\pi(\boldsymbol{\beta}) \\
&\quad \propto \exp\left\{-\frac{1}{2\sigma^2}\sum_{j=1}^{k}\sum_{i:c_i=j}(y_i - \boldsymbol{x}_i^\intercal\boldsymbol{\beta} - \boldsymbol{u}_i^\intercal\boldsymbol{\gamma})^2 - \frac{1}{2}(\boldsymbol{\beta} - \boldsymbol{b}_0)^\intercal\Sigma_0^{-1}(\boldsymbol{\beta} - \boldsymbol{b}_0)\right\} \\
&\quad = \exp\left\{-\frac{1}{2\sigma^2}\sum_{j=1}^{k}\sum_{i:c_i=j}(y_i^\gamma - \boldsymbol{x}_i^\intercal\boldsymbol{\beta})^2 - \frac{1}{2}(\boldsymbol{\beta} - \boldsymbol{b}_0)^\intercal\Sigma_0^{-1}(\boldsymbol{\beta} - \boldsymbol{b}_0)\right\}
\end{aligned}$$

where $y_i^\gamma = y_i - \boldsymbol{u}_i^\intercal\boldsymbol{\gamma}$. Therefore

$$\begin{aligned}
&\pi(\boldsymbol{\beta} \mid y_{1:n}, \boldsymbol{x}_{1:n}, \boldsymbol{u}_{1:n}, c_{1:n}, \boldsymbol{\gamma}_{1:k}, \sigma^2) \\
&\quad \propto \exp\left\{-\frac{1}{2}(\boldsymbol{\beta} - \boldsymbol{b}_n)^\intercal\Sigma_n^{-1}(\boldsymbol{\beta} - \boldsymbol{b} - n)\right\}
\end{aligned}$$

where

$$\begin{aligned}
\Sigma_n &= \left(\Sigma_0^{-1} + \frac{\mathrm{X}^\intercal\mathrm{X}}{\sigma^2}\right)^{-1}, \\
\boldsymbol{b}_n &= \Sigma_n\left(\Sigma_0^{-1}\boldsymbol{b}_0 + \frac{\mathrm{X}^\intercal(\boldsymbol{y} - \mathrm{U}\boldsymbol{\gamma})}{\sigma^2}\right).
\end{aligned}$$

Similarly, we define

$$\boldsymbol{y}_j^{\boldsymbol{\beta}} = \{y_i - \boldsymbol{x}_i^\intercal\boldsymbol{\beta} : c_i = j\}, \qquad \mathrm{U}_j^{\boldsymbol{\beta}} = \left\{\begin{bmatrix}\boldsymbol{u}_1^\intercal \\ \boldsymbol{u}_2^\intercal \\ \vdots \\ \boldsymbol{u}_{n_j}^\intercal\end{bmatrix} : c_i = j\right\},$$

and

$$\boldsymbol{\gamma}_j \mid y_{1:n}, \boldsymbol{x}_{1:n}, \boldsymbol{u}_{1:n}, c_{1:n}, \boldsymbol{\beta}, \sigma^2 \sim N(\boldsymbol{\tau}_n, \Psi_n), \qquad j = 1, \ldots, k,$$

where

$$\Psi_n = \left(\Psi_0^{-1} + \frac{(\mathrm{U}_j^{\boldsymbol{\beta}})^\intercal\mathrm{U}_j}{\sigma^2}\right)^{-1}, \qquad \boldsymbol{\tau}_n = \Psi_n\left(\Psi_0^{-1}\boldsymbol{\tau}_0 + \frac{\mathrm{U}_j^{\beta}\boldsymbol{y}_j^{\beta}}{\sigma^2}\right)^{-1}$$

□

The previous full conditional distributions can be used iteratively in a Gibbs sampler scheme to perform posterior inference. Remember, once again, that centering the random effects distribution apart from $\mathbf{0}$ can impact on the model's identifiability. Further, in presence of group-specific random intercept, we can suppress the common one. Otherwise, we can take one of the groups as baseline reference.

Exercise 1.4. Let us consider the following response variable and covariates

```
set.seed(123); betatrue <- c(-1, -2, 2);
gammatrue <- rbind(c(2, 4), c(-2, -4))
z1 <- round(rnorm(100, 0, 1), digits = 1)
z2 <- round(rnorm(100, 0, 1), digits = 1)
z3 <- round(rnorm(100, 0, 1), digits = 1)
z4 <- round(rnorm(100, 0, 1), digits = 1)
c <- rep(c(1, 2), each = 50)
X <- cbind(rep(1, 100), z1, z2)
U <- cbind(z3, z4)
tempmeans <- as.vector(X \%*\% betatrue) +
apply(cbind(U,gammatrue[c,]), 1, function(x) x[1:2] \%*\% x[3:4])
y <- sapply(tempmeans, function(x) rnorm(1, x, 1))
```

Consider a linear regression mixed model, with

$$y_i = \beta_1 + \beta_2 z_{i,1} + \beta_3 z_{i,2} + \gamma_{c_i,1} z_{i,3} + \gamma_{c_i,2} z_{i,4} + \epsilon_i.$$

with $\epsilon_i \sim N(0, \sigma^2)$.

- Implement a Gibbs sampler to perform sampling from the posterior distribution of a linear regression mixed model.
- Sample from the posterior distribution of fixed effects, random effects and variance parameters of a linear regression mixed model, with the specification given in the previous slide, assuming a priori

$$\begin{aligned} \boldsymbol{\beta} &\sim N(\mathbf{0}, \mathrm{diag}(10^3, 3)), \\ \boldsymbol{\gamma} &\sim N(\mathbf{0}, \mathrm{diag}(10^3, 2)), \\ \sigma^2 &\sim IG(2, 1). \end{aligned}$$

- Provide a first assessment of algorithm mixing and convergence.
- Provide a graphical illustration of fixed and random effect distributions, comparing random effects of different groups.

Suppose we observe covariates, but not the response variable, for a future $n+1$ observation, for which we have $z_1 = 2.2$, $z_2 = -1.1$, $z_3 = 1.4$, $z_4 = -0.7$.

- Provide a graphical illustration of the predictive distribution for the $n+1$ observation.

In the previous specification of mixed models we have considered linear models. However, mixed model can be specified in a GLM framework, to accommodate different type of re-

sponse variable and distributions, by acting on the linear predictor, as

$$\underbrace{Y_i \mid \eta_i \sim EF(b(\theta_i), \psi)}_{\text{error structure}}, \qquad \underbrace{\mu_i = g^{-1}(\eta_i)}_{\text{link function}}, \qquad \underbrace{\eta_i = \boldsymbol{x}_i^\intercal \boldsymbol{\beta} + \boldsymbol{u}_i \boldsymbol{\gamma}_{c_i}}_{\text{linear predictor}},$$

where the EF distribution can match, for example, one of the distributions presented in the previous chapter. We can handle several type of data, possibly divided into distinct groups. Usually, we do not have closed expression of the full conditional. Inference can be done resorting to STAN implementation of the model.
Finally, the same modelling structure can be extended to accommodate for multiple level of hierarchies, by including more grouping levels. This is particularly suited when we have data divided in groups, but also each group is divided in subgroups itself, and so on and so forth.

2 Model assessment

Once we estimate a model, there are some fundamental strategies to assess its bounty of fit. Ideally, we want to look at the empirical evidence supporting the model and to construct some functional to summarize such an evidence in an interpretable quantity, to compare two or more distinct models. Most of the methodologies here presented are not restricted to regression models. However, examples will relate to them.

As before, we have $y_{1:n} \in \mathbb{Y} \subseteq \mathbb{R}$ response variables that we want to model as function of $\boldsymbol{x}_{1:n} \in \mathbb{X} \subseteq \mathbb{R}^d$ suitable set of covariates (or their transformations). From a frequentist perspective, we know that if we evaluate a density function of an observation y_i at a specific value of a regression parameter $\boldsymbol{\theta}$ and possibly other parameters,

$$f(y_i \mid \boldsymbol{x}_i, \boldsymbol{\theta}),$$

we are measuring how likely is y_i given the probabilistic model assumption we are taking $f(\cdot \mid \cdot)$ and the parameter values $\boldsymbol{\theta} = \{\boldsymbol{\beta}, \dots\}$, with $\boldsymbol{\theta} \in \Theta$. Hence, the higher, the better. While combining more observations, we usually construct our dear log-likelihood function

$$\ell(y_{1:n} \mid \boldsymbol{x}_{1:n}, \boldsymbol{\theta}) = \sum_{i=1}^{n} \log f(y_i \mid \boldsymbol{x}_i, \boldsymbol{\theta}).$$

At each term of the product in the right-hand side expression, the same of above applies. Most fit measures and information criteria in the literature are based on likelihood function transformations, evaluated at a particular point estimate. For historical reasons, predictive accuracy measures are called information criteria. Typically, they are defined as function of the deviance, i.e., $-2 \log \mathcal{L}(y_{1:n} \mid \boldsymbol{x}_{1:n}, \theta) = -2\ell(y_{1:n} \mid \boldsymbol{x}_{1:n}, \boldsymbol{\theta})$.

In Bayesian inference, we have a problem. We do not have a single parameter value, but an entire distribution for such parameter. Somehow we need quantities that incorporate the entire distribution in their definition. We can resort to something that we already saw. We start from the predictive distribution of our model. For a new observation $\tilde{y}$, with covariates $\tilde{\boldsymbol{x}}$, the posterior predictive distribution is given by

$$m(\tilde{y} \mid \tilde{\boldsymbol{x}}, y_{1:n}, \boldsymbol{x}_{1:n}) = \int_{\Theta} f(\tilde{y} \mid \tilde{\boldsymbol{x}}, \boldsymbol{\theta}) \pi(\boldsymbol{\theta} \mid y_{1:n}, \boldsymbol{x}_{1:n}) \mathrm{d}\boldsymbol{\theta},$$

with $\pi(\boldsymbol{\theta} \mid y_{1:n}, \boldsymbol{x}_{1:n})$ being the posterior distribution of the parameter, given the observed sample. Hence, we solved a first issue, since with the previous quantity we do not have anymore the problem of having a random parameter.

Ideally, we can resort to the log-pointwise predictive density, by looking at the marginal distribution of each single observation given the rest, defined as

$$\begin{aligned}
\mathrm{LPPD} &= \sum_{i=1}^{n} \log m(y_i \mid y_{1:n}, \boldsymbol{x}_{1:n}) \\
&= \sum_{i=1}^{n} \log \left[\int_{\Theta} f(y_i \mid \boldsymbol{x}_i, \boldsymbol{\theta}) \pi(\boldsymbol{\theta} \mid y_{1:n}, \boldsymbol{x}_{1:n}) \mathrm{d}\boldsymbol{\theta} \right] \\
&= \sum_{i=1}^{n} \log \mathbb{E}_{\boldsymbol{\theta}} \left[f(y_i \mid \boldsymbol{x}_i, \boldsymbol{\theta}) \middle| y_{1:n}, \boldsymbol{x}_{1:n} \right].
\end{aligned}$$

The previous measures how much likely are the data under our model setting, once we marginalize the parameters. Specifically, for each observation we measure how likely it is a posteriori, averaging with respect to the posterior distribution of the parameters. We are looking at the posterior predictive performance of the estimated model. While it looks like the quantity we want to study, we unlocked a new problem, since the posterior predictive in the previous depends on the whole data. Specifically, we use the generic ith observation to both perform inference of the model, i.e. derive the posterior distribution of interest, and to compute the LPPD.

Several different criteria are available in the literature. All of them are somehow approximating the out-of-sample predictive performance. Hence, all of them have their specific flaws, but we need to measure the performance of our models. We distinguish mainly among three families.

- Within sample predictive accuracy. A rough estimate of the expected predictive distribution of new data is given by what happens with the observed data, such as the LPPD. In general, quick to evaluate and easy to interpret.
- Adjusted within sample predictive accuracy. Quantities such as AIC, DIC, WAIC are adjusted measure by a model-complexity term.
- Cross-validation. Ideally, we separate our data in train and test, then we evaluate predictive performance on a subset of data. We can obtain a synthesis of the predictive performance of our model, where observations are used either to estimate the model or to compute the predictive inference performance.

2.1 Akaike information criteria

A first index is the Akaike information criteria (AIC), one of the first index historically that has been used commonly as information criteria. Usually, inference on a parameter of interest, say $\boldsymbol{\theta}$, is summarized by a point estimate $\hat{\boldsymbol{\theta}}$, typically the maximum likelihood estimate. The AIC is defined as

$$AIC = -2\ell(y_{1:n} \mid \boldsymbol{x}_{1:n}, \hat{\boldsymbol{\theta}}_{ML}) + 2p.$$

Ideally, ignoring the multiplication by -2, we are adjusting the log-likelihood by subtracting p, the number of parameters in our model. Such a quantity mitigates the predictive performance increases we have once we include more covariates in our model specification. In

practice, the correction term adjust for overfitting. By looking at the index, we are multiplying the log-likelihood by a negative term, hence ignoring the correction, we have that for such information criteria, the smaller, the better. The correction adds a positive term. For two models having the same log-likelihood, but with different AIC values, the more complex model has a larger correction, hence resulting in higher value of AIC criteria. Hence, still the smaller, the better, since while the predictive performance are equivalent, we prefer the simpler model.

2.2 Deviance information criteria

The deviance information criteria (DIC) is, somehow, a Bayesian extension of AIC. Since we do not have a single parameter value, but an entire distribution a posteriori, we replace the maximum likelihood estimate $\hat{\boldsymbol{\theta}}_{ML}$ with the posterior mean of the parameter $\mathbb{E}[\boldsymbol{\theta} \mid y_{1:n}, \boldsymbol{x}_{1:n}]$. Similarly, the complexity penalization term p is replaced with a posterior data-based correction. Hence, we have

$$DIC = -2\ell(y_{1:n} \mid \boldsymbol{x}_{1:n}, \mathbb{E}[\boldsymbol{\theta} \mid y_{1:n}, \boldsymbol{x}_{1:n}]) + 2p_{DIC},$$

where

$$p_{DIC} = 2\left[\ell(y_{1:n} \mid \boldsymbol{x}_{1:n}, \mathbb{E}[\boldsymbol{\theta} \mid y_{1:n}, \boldsymbol{x}_{1:n}]) - \mathbb{E}_{\boldsymbol{\theta}|y_{1:n},\boldsymbol{x}_{1:n}}[\ell(y_{1:n} \mid \boldsymbol{x}_{1:n}, \boldsymbol{\theta})]\right]$$

or, equivalently,

$$\begin{aligned} p_{\mathrm{DIC}} &= 2\mathrm{Var}_{\boldsymbol{\theta}|y_{1:n},\boldsymbol{x}_{1:n}}\left(\ell(y_{1:n} \mid \boldsymbol{x}_{1:n}, \boldsymbol{\theta})\right) \\ &\approx \frac{2}{R-1}\sum_{r=1}^{R}\left[\ell(y_{1:n} \mid \boldsymbol{x}_{1:n}, \boldsymbol{\theta}^{(r)}) - \overline{\ell(y_{1:n} \mid \boldsymbol{x}_{1:n}, \boldsymbol{\theta})}\right] \end{aligned}$$

The first term in the DIC definition is still a log-likelihood, evaluated at the posterior expectation of the parameter value. Hence, also for the DIC, looking at the first term the smaller, the better. The variance term acts as a complexity penalization. For two models having the same values of the log-likelihood term, we prefer the model with the smaller values of its variance, hence resulting in more precise posterior predictive performance. For linear models with uniform priors, p_{DIC} reduces to the parameter cardinality p.

2.3 Widely applicable information criteria

The widely applicable information criteria (WAIC) somehow incorporate more posterior information rather than evaluating the log-likelihood at a specific parameter value. In practice, it is defined as a penalized version of LPPD, i.e.,

$$\mathrm{WAIC} = -2\mathrm{LPPD} + 2p_{\mathrm{WAIC}},$$

where, given a MCMC sample $\{\boldsymbol{\theta}^{(1)}, \ldots, \boldsymbol{\theta}^{(R)}\}$ from the posterior distribution, we can approximate the LPPD as

$$\mathrm{LPPD} = \sum_{i=1}^{n} \log m(y_i \mid y_{1:n}, \boldsymbol{x}_{1:n}) \approx \sum_{i=1}^{n} \log\left[\frac{1}{R}\sum_{r=1}^{R} f(y_i \mid \boldsymbol{x}_i, \boldsymbol{\theta}^{(r)})\right],$$

and the penalization term is expressed as

$$p_{WAIC} = 2\sum_{i=1}^{n}\left[\log \mathbb{E}_{\boldsymbol{\theta}|y_{1:n},\boldsymbol{x}_{1:n}}\left[f(y_i \mid \boldsymbol{x}_i, \boldsymbol{\theta})\right] - \mathbb{E}_{\boldsymbol{\theta}|y_{1:n},\boldsymbol{x}_{1:n}}\left[\log f(y_i \mid \boldsymbol{x}_i, \boldsymbol{\theta})\right]\right]$$

or equivalently

$$
\begin{aligned}
p_{\mathrm{WAIC}} &= \sum_{i=1}^{n} \mathrm{Var}_{\boldsymbol{\theta}|y_{1:n},\boldsymbol{x}_{1:n}} \left(\log f(y_i \mid \boldsymbol{x}_i, \boldsymbol{\theta})\right) \\
&\approx \sum_{i=1}^{n} \frac{1}{R-1} \sum_{r=1}^{R} \left[\log f(y_i \mid \boldsymbol{x}_i, \boldsymbol{\theta}^{(r)}) - \overline{\log f(y_i \mid \boldsymbol{x}_i, \boldsymbol{\theta})}\right]^2 .
\end{aligned}
$$

with $\overline{\log f(y_i \mid \boldsymbol{x}_i, \boldsymbol{\theta})} = \frac{1}{R}\sum_{r=1}^{R} \log f(y_i \mid \boldsymbol{x}_i, \boldsymbol{\theta}^{(r)})$. Also for the WAIC, we have a similar interpretation to the previous criteria. The first term in its definition mimics the log-likelihood term of AIC and BIC, but incorporating the parameter $\boldsymbol{\theta}$ distribution. Hence, the smaller, the better. The variance term acts as a complexity penalization, looking at how dispersed are our prediction for each observation. Whenever two models share the same LPPD value, we prefer the model with more precise predictions, i.e. the one with lower WAIC index. For large n, $\mathrm{LPML} \approx -\frac{1}{2}\mathrm{WAIC}$.

2.4 Predictive criteria

In the previous criteria, we have been optimistic by including, in each term, the same observation in both the argument of the density function and the conditioning quantities. Including the observation in the conditioning arguments may alter the prediction we are making, especially with small sample sizes. Instead of LPPD, we can consider another quantity by replacing $m(y_i \mid y_{1:n}, \boldsymbol{x}_{1:n})$ with

$$
m(y_i \mid y_{-i}, \boldsymbol{x}_{-i}) = \int_{\Theta} f(y_i \mid \boldsymbol{x}_i, \boldsymbol{\theta}) \pi(\boldsymbol{\theta} \mid y_{-i}, \boldsymbol{x}_{-i}) \mathrm{d}\boldsymbol{\theta},
$$

where y_{-i} and $\boldsymbol{x}_{-i}$ denote $y_{1:n}$ and $\boldsymbol{x}_{1:n}$ but discarding the ith element, respectively. Ideally, we want to perform a leave-one-out cross-validation. We then resort to the so-called log pseudo-marginal likelihood (LPML), which is given by

$$
\mathrm{LPML} = \sum_{i=1}^{n} \log m(y_i \mid \boldsymbol{x}_i, y_{-i}, \boldsymbol{x}_{-i}) = \sum_{i=1}^{n} \log(\mathrm{CPO}_i).
$$

where CPO_i is called the conditional predictive ordinate for observation i. We have a a new problem, as ideally we have to estimate n separate models, one for each observation removed. In practice, we can reconstruct the LPML from a generic MCMC output.

Proposition 2.1. *For a generic model with data distribution $f(y_i \mid \boldsymbol{x}_i, \boldsymbol{\theta})$ and prior distribution $\pi(\boldsymbol{\theta})$, the LPML can be estimated as*

$$
\mathrm{LPML} \approx \sum_{i=1}^{n} \log \left\{ \left[\frac{1}{R} \sum_{r=1}^{R} \frac{1}{f(y_i \mid \boldsymbol{x}_i, \boldsymbol{\theta}^{(r)})} \right]^{-1} \right\}.
$$

Proof. For a generic i, the generic CPO_i can be rewritten as

$$
\begin{aligned}
\mathrm{CPO}_i &= m(y_i \mid \boldsymbol{x}_i, y_{-i}, \boldsymbol{x}_{-i}) \\
&= \int_{\Theta} f(y_i \mid \boldsymbol{x}_i, \boldsymbol{\theta}) \frac{\prod_{\ell \neq i} f(y_\ell \mid \boldsymbol{x}_\ell, \boldsymbol{\theta})}{\int_{\Theta} \prod_{\ell \neq i} f(y_\ell \mid \boldsymbol{x}_\ell, \boldsymbol{\theta}) \pi(\boldsymbol{\theta}) \mathrm{d}\boldsymbol{\theta}} \pi(\boldsymbol{\theta}) \mathrm{d}\boldsymbol{\theta} \\
&= \frac{\int_{\Theta} \prod_{\ell=1}^{n} f(y_\ell \mid \boldsymbol{x}_\ell, \boldsymbol{\theta}) \pi(\boldsymbol{\theta}) \mathrm{d}\boldsymbol{\theta}}{\int_{\Theta} \prod_{\ell \neq i} f(y_\ell \mid \boldsymbol{x}_\ell, \boldsymbol{\theta}) \pi(\boldsymbol{\theta}) \mathrm{d}\boldsymbol{\theta}}
\end{aligned}
$$

Therefore, we have

$$
\begin{aligned}
\frac{1}{\mathrm{CPO}_i} &= \frac{\int_\Theta \prod_{\ell \neq i} f(y_\ell \mid \boldsymbol{x}_\ell, \boldsymbol{\theta})\pi(\boldsymbol{\theta})\mathrm{d}\boldsymbol{\theta}}{\int_\Theta \prod_{\ell=1}^n f(y_\ell \mid \boldsymbol{x}_\ell, \boldsymbol{\theta})\pi(\boldsymbol{\theta})\mathrm{d}\boldsymbol{\theta}} \\
&= \int_\Theta \frac{1}{f(y_i \mid \boldsymbol{x}_i, \boldsymbol{\theta})} \frac{\prod_{\ell=1}^n f(y_\ell \mid \boldsymbol{x}_\ell, \boldsymbol{\theta})\pi(\boldsymbol{\theta})}{\int_\Theta \prod_{\ell=1}^n f(y_\ell \mid \boldsymbol{x}_\ell, \boldsymbol{\theta})\pi(\boldsymbol{\theta})\mathrm{d}\boldsymbol{\theta}} \mathrm{d}\boldsymbol{\theta} \\
&= \int_\Theta \frac{1}{f(y_i \mid \boldsymbol{x}_i, \boldsymbol{\theta})} \pi(\boldsymbol{\theta} \mid y_{1:n}, \boldsymbol{x}_{1:n})\mathrm{d}\boldsymbol{\theta} \\
&= \mathbb{E}_{\boldsymbol{\theta}|y_{1:n},\boldsymbol{x}_{1:n}} \left[\frac{1}{f(y_i \mid \boldsymbol{x}_i, \boldsymbol{\theta})} \right]
\end{aligned}
$$

Hence, suppose we have a MCMC sample $\{\boldsymbol{\theta}^{(1)}, \ldots, \boldsymbol{\theta}^{(R)}\}$ from a generic algorithm. We can approximate each CPO_i as

$$
\mathrm{CPO}_i = \mathbb{E}_{\boldsymbol{\theta}|y_{1:n},\boldsymbol{x}_{1:n}} \left[\frac{1}{f(y_i \mid \boldsymbol{x}_i, \boldsymbol{\theta})} \right]^{-1} \approx \left[\frac{1}{R} \sum_{r=1}^R \frac{1}{f(y_i \mid \boldsymbol{x}_i, \boldsymbol{\theta}^{(r)})} \right]^{-1},
$$

and then the LPML becomes

$$
\mathrm{LPML} \approx \sum_{i=1}^n \log \left\{ \left[\frac{1}{R} \sum_{r=1}^R \frac{1}{f(y_i \mid \boldsymbol{x}_i, \boldsymbol{\theta}^{(r)})} \right]^{-1} \right\}.
$$

□

With the previous expression, we can easily compute the LPML by tracing by passing as output of our algorithm also the density evaluation for each observation and each iteration. In STAN we just need to include an extra line of code. Further, the LPML can be used raw as fit measure, or transformed for example introducing a model complexity penalization.

Exercise 2.2. Let us consider the following response variable and covariates

```
set.seed(123); betatrue <- c(-2, 2);
gammatrue <- rbind(c(-1, 2, 4), c(3, -2, -4))
z1 <- round(rnorm(100, 0, 1), digits = 1)
z2 <- round(rnorm(100, 0, 1), digits = 1)
z3 <- round(rnorm(100, 0, 1), digits = 1)
z4 <- round(rnorm(100, 0, 1), digits = 1)
c <- rep(c(1, 2), each = 50)
X1 <- cbind(z1, z2)
U1 <- cbind(rep(1, 100), z3, z4)
X2 <- cbind(rep(1, 100), z1, z2)
U2 <- cbind(z3, z4)
tempmeans <- as.vector(X1 \%*\% betatrue) +
apply(cbind(U1,gammatrue[c,]), 1, function(x) x[1:2] \%*\% x[3:4])
y <- sapply(tempmeans, function(x) rnorm(1, x, 1))
```

Consider the following two linear regression mixed model

$$
\begin{aligned}
LMM1: &\quad y_i = \beta_1 z_{i,1} + \beta_2 z_{i,2} + \gamma_{c_i,1} + \gamma_{c_i,2} z_{i,3} + \gamma_{c_i,3} z_{i,4} + \epsilon_i, \qquad \epsilon_i \sim N(0, \sigma^2), \\
LMM2: &\quad y_i = \beta_1 + \beta_2 z_{i,1} + \beta_3 z_{i,2} + \gamma_{c_i,1} z_{i,3} + \gamma_{c_i,2} z_{i,4} + \epsilon_i, \qquad \epsilon_i \sim N(0, \sigma^2).
\end{aligned}
$$

LMM1 has a group-specific intercept term while LMM2 has a shared intercept term.

- Make a variation of the LMM gibbs sampler of slide block 3 to return as output, for each iteration, also the density evaluation of each observation, given the current value of the parameter.
- Produce a sample from the posterior distribution of the parameters for LMM1 and LMM2 and check mixing and convergence of the algorithm, assuming a priori $\boldsymbol{\beta} \sim N(\mathbf{0}, \mathrm{diag}(10^3))$, $\boldsymbol{\gamma}_j \sim N(\mathbf{0}, \mathrm{diag}(10^3))$, $j = 1, 2$, and $\sigma^2 \sim IG(2, 2)$.
- Write the functions to calculate LPML and WAIC. Evaluate the fit measures for both the estimated model.

Consider now as third model a linear regression model

$$LM1: \qquad y_i = \beta_1 + \beta_2 z_{i,1} + \beta_3 z_{i,2} + \epsilon_i, \qquad \epsilon_i \sim N(0, \sigma^2).$$

- Write a function sampling from the posterior distribution of LM1, assuming $\boldsymbol{\beta} \sim N(\mathbf{0}, \mathrm{diag}(10^3))$ and $\sigma^2 \sim IG(2, 2)$.
- Evaluate the fit measures LPML and WAIC. Compare them with the ones obtained with LMM1 and LMM2.
- Repeat the previous two points implementing all the models in STAN.

3 COVARIATES SELECTION

Suppose we now fix a certain distribution for the data, e.g. a specific model form. Further, we have a total of p predictors available, with $\boldsymbol{x}_i^\intercal = (x_{i1}, \ldots, x_{ip})$. A reasonable question that we could ask to ourselves is whether we could find the best model for the response variable, given all the possible combination of covariates we can make. Ideally, if we can estimate all possible models combining the covariates, we can then select the best one resorting to predictive information criteria as WAIC and LPML. Each covariate can be included or not in the model. We then have 2^p possible distinct models, given p covariates. Further, when p is large, it is unfeasible to estimate all possible models and then select the best one.

We need an alternative strategy to select the best subset of covariates, without estimating all possible models. Many different regularization methods that also select covariates have been studied in the last decades. Here we consider an approach that has proven to be effective with many model strategies. We present the so called spike-and-slab approach.

Usually, the marginal distribution of each regression coefficient is diffused on a real space, hence for the generic jth coefficient β_j, we have

$$P(\beta_j = 0) = 0.$$

We augment the prior distribution of the regression coefficients defining a new prior specification which set positive probability on the 0 value (or on a neighborhood of 0). Such a prior is composed by two components:

– a spike one, concentrating the mass around 0;

– a slab one, which mimics the diffuseness of usual prior specifications.

In a general setting, we consider a model for which we have

$$g(\mathbb{E}[Y \mid \boldsymbol{x}]) = \boldsymbol{x}^\intercal \boldsymbol{\beta}, \qquad \boldsymbol{\beta} \sim \pi(\boldsymbol{\beta}).$$

Examples are the ordinary linear regression model, various GLMs, but the prior specification we are considering here can be embedded also in a mixed model environment. $\pi(\boldsymbol{\beta})$ usually is assumed to be Gaussian, but other distributions can be considered here (e.g. the Laplace distribution of the previous chapter).

At first, we consider an hierarchical specification as follows. We introduce a set of suitable augmented variables $\boldsymbol{\gamma}^\intercal = (\gamma_1, \ldots, \gamma_p)$, where each $\gamma_j \in \{0, 1\}$, $j = 1, \ldots, p$. The augmented prior can be written in a general form as

$$\pi(\boldsymbol{\beta}, \boldsymbol{\gamma}) = \pi(\boldsymbol{\beta} \mid \boldsymbol{\gamma})\pi(\boldsymbol{\gamma}).$$

Each γ_j describes the inclusion or exclusion of the jth covariate in the model, with

$$\gamma_j = \begin{cases} 1 & : \quad \text{if the } j\text{th covariate is included in the model,} \\ 0 & : \quad \text{otherwise.} \end{cases}$$

Each one of the possible 2^p models is uniquely identified by a specific realization binary sequence $\boldsymbol{\gamma}^\intercal = (\gamma_1, \ldots, \gamma_p)$. Hence, the prior is completed by setting $\gamma_j \sim Be(\theta_j)$, $j = 1, \ldots, p$, $\theta_j \sim \pi(\theta_j)$, where the latter is not mandatory but improve the model flexibility, and θ_j denotes the probability that $\boldsymbol{\beta}_j$ is large enough to be included in the model.

We have a problem here, the dimension of the parameter space change as far as we are including or excluding covariates, as well for the prior dimension. There exist suitable computational techniques to deal with this problem. However, we can consider a similar prior specification, but more tractable. Hence, instead of the augmented prior of before, we use the augmented variables γ_js to construct a mixture prior specification for each regression coefficient. We want to have a diffuse prior apart from 0, following a specific distribution. We also want to a positive probability for the coefficient of being equal (or close) to 0.

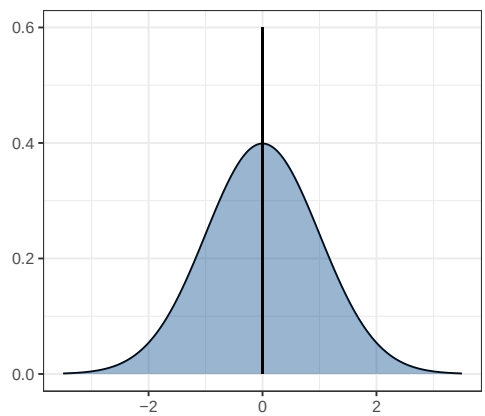

Figure 2: Graphical illustration of a spike-and-slab prior, with a component setting mass on the value $\{0\}$.

The prior specification we consider is then the following

$$\begin{aligned}
\beta_j \mid \gamma_j &\overset{ind}{\sim} (1-\gamma_j)\delta_0 + \gamma_j N(0, \tau_j^2), \\
\gamma_j \mid \theta_j &\overset{ind}{\sim} Be(\theta_j), \\
\theta_j &\overset{ind}{\sim} \pi(\theta_j),
\end{aligned}$$

for $j = 1, \ldots, p$, where δ_0 denotes a Dirac measure in 0. Such a prior is flexible to cover the 2^p possible models, while being tractable. If $\theta_j = 0.5$ for all $j = 1, \ldots, p$, we are setting a uniform prior over all the possible 2^p models. However, in practice we cannot use discrete prior distribution (such as the Dirac measure) in STAN, since it cannot handle discrete parameters.

We can consider a slightly relaxed version of the previous prior, by combining together two Gaussian distributions. A Gaussian distribution with large variance, which models actually coefficients different from 0. A Gaussian distribution with small variance, which set mass on a neighborhood of 0.

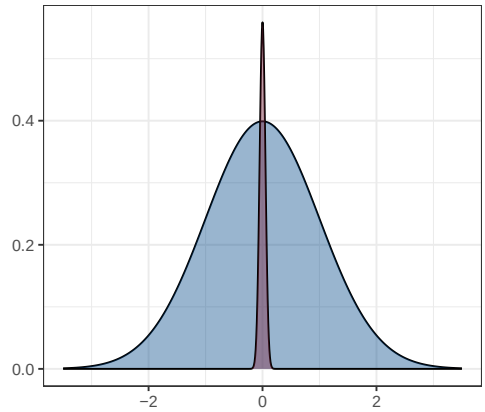

Figure 3: Graphical illustration of a more relaxed prior specification, with a component concentrating mass on a neighborhood of $\{0\}$.

The prior specification is the following

$$\begin{aligned}
\beta_j \mid \gamma_j &\overset{ind}{\sim} (1-\gamma_j) N(0, \tau_j^2) + \gamma_j N(0, c_j^2 \tau_j^2), \\
\gamma_j \mid \theta_j &\overset{ind}{\sim} Be(\theta_j), \\
\theta_j &\overset{ind}{\sim} \pi(\theta_j),
\end{aligned}$$

for $j = 1, \ldots, p$, for $\tau_j > 0$ small enough and $c_j > 0$ large enough. The mixture prior distribution for β_j is now manageable in STAN. Once we collect a sample from the posterior distribution, we can check how many times each regression coefficient has been sampled from the spike or the slab component. This approach is also called stochastic search variable selection.

A key quantity is the set of values $(-\kappa_j, \kappa_j)$ where the spike components dominates the slab components. The bounds of this set are given by $k_j = \tau_j \epsilon_j$, where

$$\epsilon_j = \sqrt{2 \frac{\log(c_j) c_j^2}{c_j^2 - 1}}.$$

When we sample a value $\beta_j \in (-\kappa_k, \kappa_k)$, we say that is close enough for zero. As already mentioned, the parameter $\boldsymbol{\gamma}^\intercal = (\gamma_1, \ldots, \gamma_p)$ uniquely identifies a specific model. Intuitively, we can compare models by looking at the specific posterior probabilities

$$P(\boldsymbol{\gamma} \mid y_{1:n}, \boldsymbol{x}_{1:n}) = \frac{m(y_{1:n} \mid \boldsymbol{x}_{1:n}, \boldsymbol{\gamma})\pi(\boldsymbol{\gamma})}{\sum_{\gamma \in \{0,1\}^p} m(y_{1:n} \mid \boldsymbol{x}_{1:n}, \boldsymbol{\gamma})\pi(\boldsymbol{\gamma})},$$

where

$$m(y_{1:n} \mid \boldsymbol{x}_{1:n}, \boldsymbol{\gamma}) = \int_{\mathbb{R}^p} \mathrm{L}(y_{1:n} \mid \boldsymbol{x}_{1:n}, \boldsymbol{\beta}, \boldsymbol{\gamma})\pi(\boldsymbol{\beta} \mid \boldsymbol{\gamma})\mathrm{d}\boldsymbol{\beta}$$

denotes the marginal distribution of the data including the covariates in $\boldsymbol{\gamma}$. Once we simulate a sample from the posterior distribution $\left(\boldsymbol{\beta}^{(r)}, \boldsymbol{\gamma}^{(r)}\right)_{r=1}^R$, we are interested into select a specific set of covariates, in force of the posterior empirical information we have. We can resort to different strategies, specifically the following.

- Highest posterior probability (HPD), we choose the model with the highest posterior probability, i.e.

$$\hat{\boldsymbol{\gamma}} = \mathrm{argmax}_{\boldsymbol{\gamma} \in \{0,1\}^p} \pi(\boldsymbol{\gamma} \mid y_{1:n}, \boldsymbol{x}_{1:n}) \approx \mathrm{argmax}_{\boldsymbol{\gamma} \in \{0,1\}^p} \frac{1}{R} \sum_{r=1}^R \mathbf{1}_{[\boldsymbol{\gamma}^{(r)} = \boldsymbol{\gamma}]},$$

  i.e. the mode of the posterior distribution.

- Median probability model (MPM), we choose the covariates having posterior probabilities of being included greater than 0.5, i.e.

$$\gamma_j \text{ such that } \pi(\gamma_j \mid y_{1:n}, \boldsymbol{x}_{1:n}) \approx \frac{1}{R} \sum_{r=1}^R \mathbf{1}_{[\gamma_j = 1]} > 0.5.$$

- Hard shrinkage (HS), we choose only the covariates that are always included in the model, i.e.

$$\gamma_j \text{ such that } \pi(\gamma_j \mid y_{1:n}, \boldsymbol{x}_{1:n}) \approx \frac{1}{R} \sum_{r=1}^R \mathbf{1}_{[\gamma_j = 1]} = 1.$$

Exercise 3.1. Consider the following data generating process

```
set.seed(123)
betatrue <- c(2, 0, 0, -3, 4, 0, -1, -2, rep(0, 12))
X <- cbind(rep(1, 100), round(matrix(rnorm(1900), ncol = 19),
           digits = 2))
tempmeans <- X \%*\% betatrue
y <- sapply(tempmeans, function(x) rnorm(1, x, 1))
```

Consider a generic linear model $y_i = \boldsymbol{x}_i^\intercal \boldsymbol{\beta} + \epsilon_i$, with $\epsilon_i \sim N(0, \sigma^2)$. We further set a priori

$$\sigma^2 \sim IG(2, 2), \qquad \beta_j \sim (1 - \theta_j)N(0, 0.001 \times 10^3) + \theta_j N(0, 10^3),\ j = 1, \ldots, p,$$

with $\theta \sim Beta(1, 1)$.

- Write the STAN code to sample from the posterior distribution of the model.
- Calculate the intersection point of the Gaussian distributions in the prior specification of β_j.
- Provide the HPD, MPM and HS estimates of the best subset of covariates for the model, and compare them with the covariates used to simulate the data.

4 Relaxing the model specification

Our model specification relies on our prior assumption, which is then updated in our posterior belief. It is possible that the prior guess we are taking it is too strongly concentrated on a specific part of the parameter space. In the worst scenario, it is strongly concentrated in the wrong part of the parameter space. In force of that, sometimes models are too sensible to the parameter specification.

A simple but effective way to avoid this problem is to include one extra hierarchical level in your model, by setting a prior distributions on the main hyperparameters of your model, ideally

$$\begin{aligned} Y_i \mid \boldsymbol{x}_i, \boldsymbol{\beta} &\sim f(y_i \mid \boldsymbol{x}_i, \boldsymbol{\beta}), \\ \boldsymbol{\beta} \mid \boldsymbol{\theta} &\sim \pi(\boldsymbol{\beta} \mid \boldsymbol{\theta}), \\ \boldsymbol{\theta} &\sim \pi(\boldsymbol{\theta}). \end{aligned}$$

Be careful to match the support of $\boldsymbol{\theta}$ with the support of the distribution assumption you are taking. In a (very) few situations, you still preserve conjugacy.

Exercise 4.1. Consider the following data generating process

```
set.seed(123); betatrue <- c(2, -3, 4, -2)
X <- cbind(rep(1, 100), round(matrix(rnorm(300), ncol = 19),
           digits = 2))
tempmeans <- X \%*\% betatrue
y <- sapply(tempmeans, function(x) rnorm(1, x, 1))
```

Consider a generic linear model $y_i = \boldsymbol{x}_i^\intercal \boldsymbol{\beta} + \epsilon_i$, with $\epsilon_i \sim N(0, \sigma^2)$.

- Write the STAN code to sample from the posterior distribution of the model, considering a priori $\sigma^2 \sim IG(100, 1), \qquad \boldsymbol{\beta} \sim N(\mathbf{25}, \mathrm{diag}(0.1))$.
- Write now the STAN code to sample from the posterior distribution of the model, considering a priori
$$\sigma^2 \sim IG(a_0, b_0), \qquad \boldsymbol{\beta} \sim N(\boldsymbol{b}_0, \Sigma_0),$$
with hyperpriors
$$\begin{aligned} a_0 &\sim Gamma(0.1, 0.1), \qquad & b_0 &\sim Gamma(0.1, 0.1), \\ \boldsymbol{b}_0 &\sim N(\mathbf{0}, \mathrm{diag}(10^3)), \qquad & \Sigma_0 &\sim Wishart(6, \mathrm{diag}(1)). \end{aligned}$$

4.1 TESTING MODELS

Once we estimate more than one model, as commonly done in data analysis, we may be interested into selecting the best model, with respect to some specific criteria. As general rule in statistical modelling, a model should be not only good in terms of fit and predictive performance, but also as simple as possible, since simple model are easy to interpret and explain. One possible strategies is to calculate for each model quantities like LPML and WAIC, then rank the model best to worst. However, even if we have a glimpse on possible predictive performance improvements, we do not see if two model significant differ from each other.

Suppose we have two distinct model, say M_1 and M_2. One first approach to select a model can be to perform comparison through tests. Hence, in a quite generic form, we have

$$\begin{aligned} M_1: &\qquad Y_i \sim f_1(y_i \mid \boldsymbol{x}_{1,i}, \boldsymbol{\theta}_1),\ i = 1, \dots, n, \qquad \boldsymbol{\theta}_1 \sim \pi_1(\boldsymbol{\theta}_1), \\ M_2: &\qquad Y_i \sim f_2(y_i \mid \boldsymbol{x}_{2,i}, \boldsymbol{\theta}_2),\ i = 1, \dots, n, \qquad \boldsymbol{\theta}_2 \sim \pi_2(\boldsymbol{\theta}_2), \end{aligned}$$

where, for example, the distributional assumption can coincide, $f_1(\cdot \mid \cdot) = f_2(\cdot \mid \cdot)$, and the model can differ only through the covariates which are including. A priori, we assume that $P(M_1) = \tau_1$ and $P(M_2) = \tau_2 = 1 - \tau_1$. The posterior probabilities of the two models can be obtained by first evaluating the marginal density for the data under M_1 and M_2, with

$$m_j(y_{1:n} \mid \boldsymbol{x}_{j,1:n}) = \int_\Theta \prod_{i=1}^n f(y_i \mid \boldsymbol{x}_i, \boldsymbol{\theta}_j)\pi_j(\boldsymbol{\theta}_j)\mathrm{d}\boldsymbol{\theta}_j, \qquad j = 1, 2.$$

Recall that the generic $m_j(y_{1:n} \mid \boldsymbol{x}_{j,1:n})$ is measuring how likely are $y_1, \dots, y_n$ under model M_j. Then, we can compute the posterior model probabilities in force of the Bayes' theorem

$$P(M_j \mid y_{1:n}, \boldsymbol{x}_{1:n}) = \frac{m(y_{1:n} \mid \boldsymbol{x}_{j,1:n}, M_j)P(M_j)}{m(y_{1:n} \mid \boldsymbol{x}_{j,1:n})} = \frac{\tau_j m_j(y_{1:n} \mid \boldsymbol{x}_{j,1:n})}{\tau_1 m_1(y_{1:n} \mid \boldsymbol{x}_{1,1:n}) + \tau_2 m_2(y_{1:n} \mid \boldsymbol{x}_{2,1:n})},$$

for $j = 1, 2$, with $P(M_2 \mid y_{1:n}, \boldsymbol{x}_{2,1:n}) = 1 - P(M_1 \mid y_{1:n}, \boldsymbol{x}_{1,1:n})$.

We can use the previous probabilities, combined with the prior ones, to test if the two model are significant different from each other, by computing the Bayes factor

$$BF_{12} = \frac{\frac{P(M_1|y_{1:n})}{P(M_2|y_{1:n})}}{\frac{P(M_1)}{P(M_2)}} = \frac{m_1(y_1, \dots, y_n)}{m_2(y_1, \dots, y_n)}.$$

But what if we have more than two models? A possible strategy is to consider jointly the posterior probabilities of different models. Suppose we have $k \geq 2$ models $M_1, \dots, M_k$. A priori, without any further information, we set the models equally probable

$$P(M_j) = \frac{1}{k}, \qquad j = 1, \dots, k.$$

Similarly to before, we have different distributional assumptions $\{f_j(y_i \mid \boldsymbol{x}_{j,i}, \boldsymbol{\beta}_j)\}_{j=1}^k$ and prior assumptions $\pi_j(\boldsymbol{\beta}_j)$ that lead to different marginal distributions $\{m_j(y_i \mid \boldsymbol{x}_{j,1:n})\}_{j=1}^k$. Hence, we can compute the posterior probability of each model as

$$P(M_j \mid y_{1:n}, \boldsymbol{x}_{j,1:n}) = \frac{m_j(y_{1:n} \mid \boldsymbol{x}_{j,1:n})P(M_j)}{m(y_{1:n} \mid \boldsymbol{x}_{1:n})},$$

where $m(y_{1:n} \mid \boldsymbol{x}_{1:n}) = \sum_{j=1}^k m_j(y_{1:n} \mid \boldsymbol{x}_{j,1:n})P(M_j)$. One can then chose, for example, the model which has the largest posterior probability $P(M_j \mid y_{1:n}, \boldsymbol{x}_{j,1:n})$.

Exercise 4.2. Let us consider the following response variable and covariates

```
set.seed(123); betatrue <- c(-4, 2, -4, 0)
z1 <- round(rnorm(100, 0, 1), digits = 1)
z2 <- round(rnorm(100, 0, 1), digits = 1)
z3 <- round(rnorm(100, 0, 1), digits = 1)
X <- cbind(rep(1, 100), z1, z2, z3)
tempprobs <- exp(as.vector(X \%*\% betatrue)) /
                (1 + exp(as.vector(X \%*\% betatrue)))
y <- sapply(tempprobs[,1], function(x) rbinom(1,1,x))
```

Consider three distinct logistic models, specifically

$$
\begin{aligned}
&M_1 : y_i \sim Be(\theta_i), \qquad \theta_i = \text{logit}(\beta_1 + \beta_2 z_{i,1} + \beta_3 z_{i,2} + \beta_4 z_{i,3}), \qquad i = 1, \ldots, n,\\
&M_2 : y_i \sim Be(\theta_i), \qquad \theta_i = \text{logit}(\beta_1 + \beta_2 z_{i,1} + \beta_3 z_{i,2}), \qquad i = 1, \ldots, n,\\
&M_3 : y_i \sim Be(\theta_i + \beta_2 z_{i,1}), \qquad \theta_i = \text{logit}(\beta_1), \qquad i = 1, \ldots, n.
\end{aligned}
$$

- Write the augmented Gibbs sampler for the logistic model, which is also returning the pmf value of each observation.
- Select the model which maximize the posterior probability.
- Test the selected model versus the simpler one.
- Repeat the previous points in STAN.

5 Bayesian model averaging

Suppose we are estimating more than one model, say $M_1, \ldots, M_m$. Instead of selecting a single model, resorting to information criteria or Bayesian testing, we can combine the models together, incorporating all the posterior information we have. For each model a priori we have a specific prior probability, here denoted as $P(M_1), \ldots, P(M_m)$. Suppose that for each model we have a prior probability distribution, i.e., for M_j we consider a prior $\pi_j(\boldsymbol{\theta}_j)$, where $\boldsymbol{\theta}_j$ is the parameter vector indexing the jth model. A posteriori we have, for the generic jth mdoel,

$$
P(M_j \mid y_{1:n}, \boldsymbol{x}_{1:n}) = \frac{m(y_{1:n} \mid \boldsymbol{x}_{j,1:n}, M_j) P(M_j)}{\sum_{j=1}^m m(y_{1:n} \mid \boldsymbol{x}_{j,1:n}, M_j) P(M_j)},
$$

where

$$
m(y_{1:n} \mid \boldsymbol{x}_{j,1:n}, M_j) = \int_{\Theta_j} f_j(y_i \mid \boldsymbol{x}_i, \boldsymbol{\theta}_j) \pi(\boldsymbol{\theta}_j) \mathrm{d}\boldsymbol{\theta}_j.
$$

Note that the data distribution f_j, the parameter vector $\boldsymbol{\theta}_j$, the prior distribution $\pi_j(\boldsymbol{\theta}_j)$ can be possibly model-specific.

In predictive inference, a posteriori we are usually interested into estimating the distribution of a single observation, say Y_{n+1}, or a quantity of interest, either a functional of Y_{n+1} distribution or some parameter of interest. In the Bayesian model averaging such quantities are obtained as weighted averages of all the possible models, where the weights are the

posterior model probabilities $P(M_j \mid y_{1:n}, \boldsymbol{x}_{1:n})$. Suppose that the quantity of interest is Δ. Hence, the Bayesian model averaging estimate for Δ is given by

$$\mathbb{E}[\Delta \mid y_{1:n}, \boldsymbol{x}_{1:n}] = \sum_{j=1}^{m} \mathbb{E}[\Delta \mid M_j, y_{1:n}, \boldsymbol{x}_{1:n}] P(M_j \mid y_{1:n}, \boldsymbol{x}_{1:n}).$$

The previous model averaging yield the following variability

$$\begin{aligned}\mathbb{V}\text{ar}(\Delta \mid y_{1:n}, \boldsymbol{x}_{1:n}) &= \sum_{j=1}^{m} P(M_j \mid y_{1:n}, \boldsymbol{x}_{1:n}) \mathbb{V}\text{ar}(\Delta \mid M_j, y_{1:n}, \boldsymbol{x}_{1:n}) \\ &\quad + \sum_{j=1}^{m} P(M_j \mid y_{1:n}, \boldsymbol{x}_{1:n}) \Big(\mathbb{E}[\Delta \mid M_j, y_{1:n}, \boldsymbol{x}_{1:n}] - \mathbb{E}[\Delta \mid y_{1:n}, \boldsymbol{x}_{1:n}] \Big)^2.\end{aligned}$$

The first term in the previous variance is the weighted average of the model specific variances. The second term, instead, look at how much the models agree on estimating the quantity Δ. If the models have the same expectation for Δ, then the second term cancel out. Some remarkable example are the followings.

- Regression coefficients with all the possible models. If we consider a model with p covariates, we have 2^p possible models obtained by including or excluding each single covariate. Hence, the quantity of interest Δ is the vector of regression coefficients $\boldsymbol{\beta}$, the prior is model-specific since it include some of the covariates, and the data distribution is shared across the models. A posteriori, we have

  $$\mathbb{E}[\boldsymbol{\beta} \mid \boldsymbol{y}_{1:n}, \boldsymbol{x}_{1:n}] = \sum_{m=1}^{2^p} P(M_j \mid y_{1:n}, \boldsymbol{x}_{1:n}) \mathbb{E}[\boldsymbol{\beta} \mid M_j, y_{1:n}, \boldsymbol{x}_{1:n}],$$

  where $\beta_i \mid M_j = 0$ if the model j does not include the ith covariate.

- Combining predictive distribution with different regularization, we are interested into predictive inference for the $n+1$ observation, given $y_{1:n}$ and $\boldsymbol{x}_{1:n}$, but we want to average different prior specifications inducing different regularizations on the regression coefficients, e.g. $\pi_1(\boldsymbol{\beta})$ a Gaussian distribution and $\pi_2(\boldsymbol{\beta})$ a product of independent Laplace distributions. The quantity of interest Δ is the marginal distribution $m(y_{n+1} \mid \boldsymbol{x}_{n+1}, y_{1:n}, \boldsymbol{x}_{1:n})$ of the $n+1$ observation. Hence, we have

  $$m(y_{n+1} \mid \boldsymbol{x}_{n+1}, y_{1:n}, \boldsymbol{x}_{1:n}) = \sum_{j=1}^{2} P(M_j \mid y_{1:n}, \boldsymbol{x}_{1:n}) m(y_{n+1} \mid M_j, \boldsymbol{x}_{n+1}, y_{1:n}, \boldsymbol{x}_{1:n})$$

  where

  $$m(y_{n+1} \mid M_j, \boldsymbol{x}_{n+1}, y_{1:n}, \boldsymbol{x}_{1:n}) = \mathbb{E}_{\boldsymbol{\beta}} \Big[f(y_{n+1} \mid \boldsymbol{x}_{n+1}, \boldsymbol{\beta}) \mid M_j, y_{1:n}, \boldsymbol{x}_{1:n} \Big]$$

  is the marginal distribution of the jth model.